Jack E. Timmermans

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EDUCATION

Boston College, **Morrissey College of Arts and Sciences** Chestnut Hill, MA
Bachelor of Science in Mathematics and Computer Science, Double Major May 2027
Overall GPA: 3.94, Dean's List First Honors (all semesters)
Relevant Coursework: Analysis I & II, Linear Algebra (Honors), Multivariable Calculus (Honors), Probability, Mathematical Statistics, Machine Learning, Algorithms, Natural Language Processing, Advanced Data Analysis, Computer Networks, Economics and Computation

RESEARCH EXPERIENCE

Undergraduate Research Fellowship, Department of Computer Science, Boston College

Quantized Model Surrogate Loss Guarantees *(Advisor: Professor J. Finocchiaro)* June 2026 - Present

- Developed theoretical analyses surrounding surrogate loss function guarantees for model classes after post-training quantization on model weights.
- Leveraged existing notions of H-consistency and loss-regret transfer that establish when surrogate losses effectively approximate target losses for a given model class.
- Derived counterexamples characterizing conditions for model consistency under quantization and broader H-consistency given distributional assumptions, computationally validating these examples by optimizing surrogate loss functions over model restricted classes with PyTorch.

Autoencoders for Missing Person Detection Project *(External collaboration)* Aug - Sep 2026

- Developed autoencoder models for unsupervised anomaly detection in open-water satellite imagery, investigating potential for detecting anomalous regions associated with missing persons.
- Designed and evaluated experimental approaches using spatial clustering metrics, including silhouette score, and anomaly-score thresholding methods such as ISODATA to assess and improve detection performance.

Optimal Ridge Regularization Revisited *(Advisor: Professor S. Alvarez)* Apr 2025 - May 2026

- Developed a fixed-point iterative algorithm leveraging a decomposition of the input data to select a near-optimal regularization parameter for ridge regression.
- Proved the contractiveness of the fixed-point mapping, establishing the theoretical validity and convergence of the procedure.
- Designed and conducted synthetic-data experiments across varying sample sizes, aspect ratios, and noise levels, demonstrating that the procedure achieves near-optimal generalization performance.

PUBLICATIONS

Timmermans, J., Alvarez, S. A. *Optimal Ridge Regularization Revisited.* Manuscript under review at *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.

TECHNICAL SKILLS

Machine Learning & Data Science: Python (NumPy, Pandas, Matplotlib, scikit-learn, PyTorch), Statistical Modeling, Experimental Design
Programming & Development: Java, C, HTML, Git, Go, CSS, JavaScript
Tools & Visualization: LaTeX, Blender, Unreal Engine
Language: Spanish (Proficient)

ADDITIONAL EXPERIENCE

Tutor, *Julie's Family Learning Program*, Boston, MA Sept 2024 - May 2025

- Tutored adult learners in reading, writing, technology, mathematics, and qualitative reasoning to support preparation for high school credentials.